

EVOLVEWEB AI – AI POWERED WEBSITE BUILDER

Gagan Kumar Giri (11222690) | Ishika Bansal (11222567)

Supervisor: Ms. Arpneek Kaur, Assistant Professor | MMEC, Mullana | April 2026



ABSTRACT

EvolveWeb AI is an AI-powered web application that fully automates website creation from a plain-language idea description. Users describe their startup concept and the system generates a professional, responsive, production-ready website in minutes — no coding required. Built on the MERN stack with OpenRouter/DeepSeek AI integration.

PROBLEM STATEMENT

- Website development needs technical skills, weeks of effort, and high cost.
- Students, startups & SMEs cannot afford professional developers.
- Existing builders (Wix, Webflow) are template-based with no real AI automation.
- No platform offers a full idea-to-code pipeline in one authenticated system.

OBJECTIVES

- Generate complete websites from natural language idea descriptions.
- Implement a 5-stage AI pipeline: Analysis → Evolution → Blueprint → Structure → Code.
- Provide secure JWT authentication and full project management.
- Reduce development time from weeks to under 5 minutes.
- Enable live preview and ZIP export of generated websites.

TECHNOLOGY STACK

Layer	Technology
Frontend	React.js, Tailwind CSS, Framer Motion
Backend	Node.js, Express.js REST API
Database	MongoDB Atlas (Mongoose)
Auth	JWT + bcryptjs
AI Model	DeepSeek via OpenRouter API
Export	Archiver (ZIP) + Blob URL Preview

SYSTEM ARCHITECTURE

4-layer architecture communicating via RESTful APIs, secured by JWT throughout:

USER (Web Browser)
Frontend – React.js + Tailwind + Framer Motion
Backend API – Node.js + Express.js + JWT
AI – OpenRouter API / DeepSeek Model
Database – MongoDB Atlas

5-STAGE AI PIPELINE

1	Idea Analysis – Score + SWOT
2	Idea Evolution – Refine concept
3	Blueprint – USP, Features, Target
4	Structure – Multi-page architecture
5	Code Gen – Full HTML/CSS/JS site

KEY FEATURES

- AI multi-stage website generation pipeline
- Live preview (sandboxed iframe via Blob URL)
- ZIP project export via Archiver library
- JWT auth + bcrypt password hashing
- Animated dashboard with Framer Motion
- Responsive UI – mobile & desktop
- Full project CRUD management dashboard

PLATFORM REQUIREMENTS

Hardware: Min. 4GB RAM, 2GHz processor, internet, browser

Frontend: React.js, Tailwind CSS, Framer Motion, React Router

Backend: Node.js, Express.js, OpenAI SDK (OpenRouter)

Database: MongoDB Atlas, Mongoose ODM

Auth/Sec: JWT, bcryptjs

Dev Tools: VS Code, GitHub, Vite, Nodemon, Postman

RESULTS

All 5 pipeline stages work correctly. The system generates complete, responsive websites from idea descriptions in under 5 minutes. JWT auth, project storage, live preview, and ZIP export all function as designed. All objectives successfully achieved.

CHALLENGES FACED

- AI JSON Parsing: DeepSeek occasionally returned malformed JSON — solved with a robust `parseAIResponse()` cleanup function.
- Live Preview Security: Rendering AI-generated HTML safely — solved using sandboxed iframe with Blob URLs.
- Pipeline Consistency: Ensuring all 5 stages produced usable output — solved with structured prompts and fallback handlers.

FUTURE SCOPE

- One-click deployment to Vercel / Netlify / AWS
- Drag-and-drop visual customization builder
- SEO optimization & multi-language support
- Template marketplace & SaaS billing (Stripe)
- Team collaboration features

CONCLUSION

EvolveWeb AI demonstrates that a 5-stage AI pipeline on a MERN stack can transform a plain-language idea into a production-ready website, democratizing web development for students, startups, and non-technical users.

References: react.dev | nodejs.org | mongodb.com | expressjs.com | jwt.io | openrouter.ai | deepseek.com | tailwindcss.com | framer.com/motion | ieeeexplore.ieee.org